Competency 12: Explores applicability of ICT to business organizations and the competitive marketplace

Competency Level 12.1: Explores the role of ICT in the world of business

Learning Outcomes:

- Defines digital economy
- Lists and describes new business methods in digital economy
- Identifies the concepts behind pure brick, brick and click, and pure click organizations
- Describes the role of ICT in business functions of an organization

E-Commerce

ICT, Business and Competitiveness

Role of ICT in Modern Business

The relationship between Information and Communication Technology (ICT) and businesses is ever-growing. For few decades, ICT has been playing the role of the key enabler for successful businesses. In other words, ICT enables providing solutions to various business problems, especially such as inefficiencies in business processes, lack of productivity, challenging business competition, ineffective business models, increasing expectations of customers, inability to reach wider markets, etc. It all started with ICT being successfully deployed in computerizing the manual business processes of companies a few decades ago. Since then, the ICT kept on advancing every year and no public or private company nowadays can think of business success excluding ICT from the picture.

Digital Economy

The digital economy is an economy based on digital computing technologies[1]. In other words, economic activities in a digital economy happen through digital mediums. Economies of most countries are now transformed into digital economies. Think about the way people read newspapers nowadays. Many people in Sri Lanka, especially the younger generation, do not buy printed newspapers. Instead, they read newspapers online. Not only that; they provide instant feedback on articles they read, share them among friends on social media and conduct discussions based on them. This scenario reflects a very important development which is a result of the advancement of ICT. That is, the advancement of ICT has created a platform for people to engage in economic activities in digital form. For example, people can search for products online, visually inspect their features in 2D and 3D form and purchase products online by making payments using digital or other virtual currencies. Another example of this is, a student can take up short-term freelancing jobs offered by companies in other countries such as translation jobs online, as part-time work to support their education. Thus, notably, the world's economy is more and more transforming to a digital economy with the advancement of ICT in which the production, distribution, and consumption of goods and services happening in digital form.

[1] This should not be confused with the subject you learn as Economics. Students should focus on the term 'Digital Economy'.

What advancement, in particular, has made this transformation? It is Internet technology. With Internet technology, especially the Web 2.0 technology, we can see a huge array of digital products such as product catalogs, TV and Radio programs, video content, music, books, magazines, newspaper articles, software and many more being produced, distributed and consumed worldwide. The internet technology has also given rise to the use of virtual currencies such as cryptocurrencies[2] making alternative digital forms of payments available for merchants and consumers. The internet technology has further made it possible nowadays to connect people with people but also people with things as well as things with things! For example, home appliances such as washing machines and refrigerators come with embedded microcontrollers and networking capabilities so that they could be connected via the Internet with human users as well as other appliances (and systems) alike. These capabilities made possible by the advancing Internet technology have reduced the geographical, social and cultural distances between firms and people around the world, making the entire world one global village. This is what we call globalization.

ICT and Business Transformation

ICT, as the key enabler of modern business success, transforms the way companies operate. With more and more people browsing the Internet, especially using mobile and handheld devices, and with more and more people use social networks, organizations have equal challenges and opportunities in the market. When people are exposed to more and more information through the Internet about products and services, their expectations as customers elevate rapidly. On the other hand, due to globalization, business competition among firms too tends to grow as international market players could enter into markets avoiding geographical barriers with low-cost substitute products as well as products with world-class technologies. However, wiser organizations transform themselves strategically using ICT effectively to explore more and more opportunities in the digital economy while facing the challenges arisen by the same.

The key ICT-based tool for business transformation is the *information system.* Organizations use information systems to strategically transform themselves to get a competitive advantage. That means information systems help organizations to implement a business strategy to operate better than its competitors in the market. An ICT-based information system uses computing technologies to collect, store and disseminate business data and, process them to get useful information and make them available for key stakeholders of the firm (such as managers, suppliers, and customers) to make decisions. Particularly, information systems help organizations to transform themselves into *digital firms* in which the key business processes are accomplished through digital networks, key business relationships are digitally enabled and mediated and the key business assets are digitally managed[3]. For example, Information systems help placing and handling orders as well as maintaining relationships with suppliers electronically letting them to view internal information such as production schedules and getting the information transferred via electronic data interchange (EDI).

ICT-enabled digital firms can transform themselves with new business models as alternatives for traditional business models. A *business model* is a particular way of doing business. *Pure brick* organizations, which are usually selling physical products offline are typically following a traditional business model where the customers visit the store, talk to sales staff, place orders at sales counters, make cash payments (unless corporate customers to whom some credit period

[2] Read about bitcoins and other similar cryptocurrencies being used in online business

[3] Read Chapter 01 of Management Information Systems by Laudon and Laudon (12E or later) for more details

will be given), produce the invoice to the warehouse, collect items and leave. However, with the advancement of ICT, organizations can become *pure click* organizations as well as *brick and click* organizations and come up with innovative business models which strategically help them to get a competitive advantage. For example, most modern insurance companies and banks have become brick and click organizations with innovative business models that make use of the Internet. That means, such organizations have introduced an online business component in addition to the traditional offline business component. For example, in the case of motor insurance claiming, apart from the traditional model of motor insurance claiming some insurance firms offer to click to claim facility where the driver can contact the call center over the phone, proceeding the details of an accident, uploading photos of the necessary documents as well as the details damages through a mobile app and leave the scene after a verification process[4].

Some other good examples of new business models are reverse auctions, group purchasing, and e-marketplaces. In reverse auctions, traditional roles of the buyer and seller are reversed. Instead of buyers who compete for goods by making bids, in reverse auctions sellers compete to sell the product by placing bids in response to a product or service request placed by a single buyer[5]. Group purchasing, on the other hand, helps making use of economies of scale. A seller can offer goods at a price lower than the set price and still make a good profit if the quantity sold is larger. Group purchasing makes use of this opportunity by getting orders from several individuals, aggregating them into a single order (as one order from a group of buyers) and placing it with a seller who is willing to fulfill that order. E-market places are virtual platforms (online platforms) on which buyers and sellers can meet. In such e-marketplaces, sellers can open virtual stores and buyers can search for goods and make purchases online[6]. Notably, these three business models are successfully being used by pure click organizations which entirely exist on the Internet. Such organizations do not have physical premises and they do not engage in offline business as well. With well-set communication, payment and distribution infrastructure in place (thankful to ICT), pure click organizations ensure that the right item is delivered to the right customer at right time in the right quality.

ICT Applications in Business Functions

ICT applications in business are diverse. On one hand, it could be in diverse industries including health, education, agriculture, banking and hospitality. On the other hand, it could be in business functions such as accounting, human resource management, production, sales, and marketing. Other than that, there are common ICT applications in supply chain management (SCM), customer relationship management (CRM) and business communication. In any case, however, ICT application involves building an information system which helps converting data to useful information and thereby helps managing a particular business process efficiently and effectively. Furthermore, advancing the Internet of things (IoT) and mobile computing technologies as well as the pervasive use of smartphones enables smart systems which supports the foregoing of many industries. For example, smart classrooms, smart health systems, and smart agricultural systems extensively make use of such technologies to fulfill various objectives in respective industries.

[4] Read about Click2Claim of fairfirst Insurance, Sri Lanka.

[5] Watch the video https://www.youtube.com/watch?v=vdqPHgGKgjU for more details.

[6] Visit www.ebay.com and see how an e-marketplace work. Note that eBay is also an example for a forward auction.

Use of ICT in business functions as information systems could be broadly classified into two categories; transaction processing and decision support. Transactions are elementary activities being recorded in organizations. For example, a sale taking place at a supermarket is a transaction. Similarly, a new delivery of stocks, recruitment of a new employee, a cash withdrawal at an ATM by a customer, a book borrowed at a library and a new item manufactured at a factory are examples for transactions. Every business function needs information systems to record such transactions for future use. Typically, transaction processing systems help keeping track of such transactions in business functions. On the other hand, managers need information systems to help them to make decisions. For example, at the end of the day, some managers may need to know how much raw material stock is remaining to decide whether to make a raw material order or not. Another group of managers may want to forecast sales in the year to come by simulating sales patterns according to different market conditions. The most senior managers would need to know the overall health of the organizations to make strategies for a future couple of years. Organizations use management information systems (MIS), decision support systems (DSS) and executive support systems (ESS) for these purposes.

ICT enabled information systems are being used in every business function with unique objectives and characteristics.

- **Accounting:** According to the business dictionary[7], accounting is the practice and body of knowledge concerned primarily with methods for recording transactions, keeping financial records, performing internal audits, reporting and analyzing financial information to the management and advising on taxation matters. Thus, it is the systematic process of dealing with incomes, expenditures, assets, and liabilities of an organization. For example, recording transactions pertaining to incomes and expenditures of a given period helps calculating the overall profit or loss during that period. Accounting information systems help performing various accounting activities with the help of ICT[8]. For example, handling accounts payable is an activity of the Accounts Department. It will handle payments to be made to the suppliers for items they supplied to the company such as raw materials. There, the system helps to keep track of purchase orders made to suppliers, inventory records on goods actually received (according to purchase orders) and the invoices received from suppliers pertaining to the purchase orders they completed. The system also helps to cross-check the purchase order details, inventory records and the details on the invoices and verify before authorizing payments. This could have been a tedious task if it was done manually but the ICT-based information systems make it accurate, efficient and fast.
- **Human Resources Management (HRM):** Human resource management function deals with managing people of the organization. Activities of the HRM function includerecruitment, transfer, training, evaluation, maintaining attendance, etc. HRM function to utilizes ICT-based information systems for managing these activities. In fact, in modern human resource information systems (HRIS), there exist modules, which mean sub-systems, to deal with those activities separately, For example, the recruitment module will help to manage the recruitment process by keeping (transaction) data related to managing CVs, screening for interviews, interview results, recruitments, etc. Apart from these, some organizations provide self-service systems to employees to access their data related to salary, deductions, leaves, and attendance. This is typically done through a web portal in which the employees login to the system through a web interface. Once logged into the portal, employees can see the services made available to them through the system.

[7] Read www.businessdictionary.com/definition/accounting.html for more details.

[8] Accounting information systems heavily rely on data from other functions. See https://www.slideshare.net/wiweck/accounting-information-system-18527651 for details.

- **Production:** According to Laudon and Laudon, production management is concerned with the planning and control of all those activities that transform inputs into outputs by adding some value to inputs. For example, in a soft drinks production plant, inputs such as sugar, flavors, water, etc. will be converted into soft drinks and come out in containers (bottles) as the output. In such a plant, the managers have to plan what to produce, when, in what quantities and where to distribute. If they produce the wrong product, that will be rejected by the consumers. If they produce more, excess stock will be remaining in the warehouses. If they produce less, there will be shortages in the market. This will make the consumers unhappy. Thus, there exist standard activities in production management such as product design, development, production planning, scheduling, etc. All these activities require data and, once again ICT-based information systems play a critical role in assisting managers to manage the production.
- **Sales and Marketing:** No business can survive without winning the hearts of the customers. All companies try to make the right product available to customers through the right channel at the right price. Therefore, all successful firms attempt to understand their customers better and try to keep them satisfied and loyal to the company. ICT provides means for companies to understand their customers by segmenting them based on purchase data, knowing what items are usually bought together and also by predicting who will buy what, when and where. Furthermore, Web 2.0 technology, as well as the resulting social media platforms, has enabled the companies to listen to their customers in the form of ratings, feedbacks, status updates, blogs, comments, tags, etc. Analysis of these different types of data gives rich market insights to firms to compete better in the market.
- **Supply Chain Management:** Supply chain links a firm with its external partners such as raw material suppliers, logistic companies and distributors. However, if the activities of these partners are not synchronized with the production activities of the firm, the end products cannot be successfully offered to the market. ICT-based supply chain management systems[9] play a critical role here by automating the flow of information across organizational boundaries and thereby linking external actors in the supply chain with the firm. For example, supplier organizations can view production schedules of the firm (by connecting to the production management information system through a web interface) and accordingly adjust their production schedule, so that an uninterrupted row material supply is ensured.
- **Business Communication:** ICT has changed the way communication happens in organizations. E_mail had been a popular mode of business communication fora couple of decades. With the development of broadband technologies, communication platforms such as Skype and Zoom became popular for video conferencing between employees and managers in geographically dispersed locations. Moreover, the broadband Internet, as well as the pervasive use of mobile devices, has given rise to more mobile-based communication platforms such as Viber and WhatsApp. These mobile apps help taking voice and video, sharing documents, pictures, videos and other files, as well as instant messaging. Due to these changes in business communication enabled by ICT, it has even made it possible for employees to work from anywhere (teleworking, telecommuting)which is getting increasingly popular especially among people like freelancers and working mothers.

Moreover, emerging technologies such as the Internet of Things (IoT), Big Data and Cloud Computing together with social computing has paved way for smarter systems to be increasingly

[9] Search and read about the term inter-organizational systems on Google.

applied in the context of business management and communication. For example, paradigms such as Industry 4.0[10] focuses on creating cyber-physical systems of automation and data exchange in manufacturing environments with the help of sensors, Internet-based communication links, and big data analytics. Smart systems are getting popular in industries such as agriculture, healthcare, education, logistics as well as hospitality. For example, smart classrooms facilitate value-added learning for the learners whereas tour guide systems supported by augmented reality help tourists to have better experiences worldwide[11].

[10] Read the article on https://www.baslerweb.com/en/vision-campus/markets-and-applications/image-processing-industry-4-0/?gclid=EAIaIQobChMI9sDOv8S62gIV1iMrCh0SgwgqEAAYAiAAEgKUIvD_BwE

[11] Read the article on https://thinkdigital.travel/wp-content/uploads/2013/04/10-AR-Best-Practices-in-Tourism.pdf

Competency Level 12.2 Analyses the relationship between ICT and business operations

Learning Outcomes:

- Distinguishes the e-commerce and e-business
- Investigates the scope of e -commerce and e- business
- Lists and briefly describes the types of e- commerce transactions
- Identifies the advantages and disadvantages of e- commerce

Electronic Business

Electronic business (E-business) could be simply explained as doing business electronically. This means that the business processes are conducted with the help of a computer network based on Internet technology. It helps data and information to be shared quickly within and across business functions, as well as external parties, and thereby the business processes to be conducted effectively and efficiently. As most early security concerned pertaining to the use of the Internet have now been addressed, we could see major organizations re-thinking their businesses in terms of the Internet and conducting e-business to buy parts and supplies from other companies, collaborate on sales promotions, and conduct joint research[12].

E-Business Operation

Most students get confused with the two terms of*electronic commerce (e-commerce)* and *electronic business*. Compared to e-commerce, e-business is a broader concept. E-commerce is actually a part of e-business. Broadly speaking e-business facilitates business transactions to be conducted electronically. Think about the order fulfillment process of a company which happens electronically. It is actually a cross-functional process which involves the sales, production, finance, and warehouse (and distribution) functions of the organization. Therefore, there exists an information system module (i.e. an information system component) for the business activities of each function. However, as shown in ***figure 4.2.1*** these modules are integrated as a single unit by connecting them to a single database through internet technology.

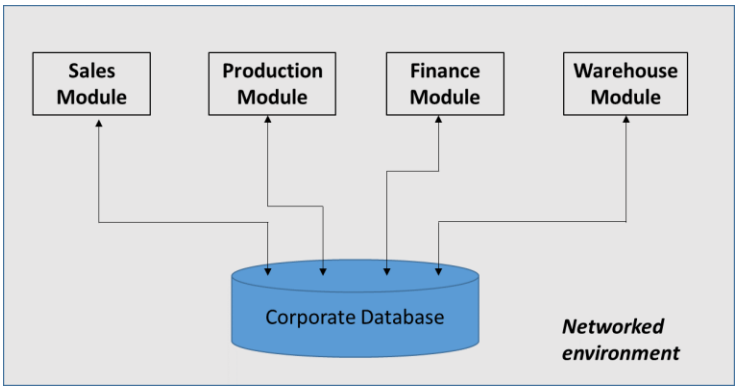

Figure 4.2.1: Integrated modules in an e-business environment

With this architecture, details of transactions of one function become accessible to the other functions and even for external parties[13]. For example, a purchase order generated through the sales module will be saved in the database. The production module, as well as the finance module, will simultaneously get an alert of the new purchase order added into the system. Based on that, the production staff will view the corresponding items to be produced and update

[12] Read the article https://searchcio.techtarget.com/definition/e-business

[13] There involves security measures to avoid information getting into unnecessary hands.

production schedules accordingly through the production module. At the same time, the finance staff will generate the invoice according to the requested items in the purchase order as they too can view the items through the finance module. Once the production is completed the status of the purchase order will be updated through the production module as completed and the warehouse staff will accordingly be alerted. The warehouse staff will then arrange the shipment once the payment is done according to the invoice generated earlier. Notably in this scenario, the electronic form of transactions enables better coordination among business functions and reduces the chances of errors and delays taking place in the process. Furthermore, access to the central corporate database could be provided to external parties such as customers, suppliers, and distributors through appropriate modules and interfaces. For example, when you access your bank account through an e-banking system and do online transactions, you actually perform e-business transactions[14]. Practically, organizations adopt e-business capabilities through enterprise resource planning (ERP) systems which has a broader set of standard modules for various functions of a typical organization[15].

Electronic Commerce

Electronic commerce (e-commerce) is simply a part of electronic business. E-commerce performs buying and selling activities of products and services online. For example, an e-commerce transaction takes place when you buy a book from Amazon over the Internet. Similarly, when you transfer money from your account to someone else's account using your bank's e-banking service, an e-commerce transaction takes place. The difference, in this case, is the transaction involves a service, not a product. E-commerce could be seen as a system of various components and actors that act and interact to enable online transactions to happen to exchange goods and services. Those factors include online marketplaces, virtual storefronts, information brokers, content providers, portals, online service providers and virtual communities.

- **Online marketplace:** Online marketplace is an e-commerce site which acts as a platform for multiple large and small-scale sellers to offer their products and services to a global pool of customers. For example, individual merchants can open virtual stores on Amazon's platform (Amazon's online marketplace) and sell their products and services. Rover.com is another interesting online marketplace on which a range of products and services related to pet care is being offered[16].

- **Virtual storefronts**: Virtual storefronts make use of digital technologies to display products being offered on an electronic screen. Tesco is a famous company for opening virtual storefronts in places like airports and subway stations. Figure4.2.2[17]shows a virtual storefront opened by Tesco at an airport and a customer adding items on display to a shopping cart using a mobile app by scanning the QR code associated with each item. Through the app, the customer can confirm her order and Tesco delivers the goods to the intended destination. On the other hand, all online shops available in online market places are examples of virtual storefronts.

[14] E-banking is actually e-commerce but it is a part of e-business.

[15] Read the article http://www.netsuite.com/portal/resource/articles/erp/what-is-erp.shtml

[16] Read the article https://hackernoon.com/top-5-most-popular-online-marketplaces-how-to-join-the-champions-league-a313dbdfd338

[17] Figure source 4.2.2: http://vonbismark.com/wp-content/uploads/2012/08/tesco-trials-interactive-virtual-store-gatwick-0.jpg

Figure4.2.2: A customer purchasing from a virtual storefront

- **Information brokers:** Information brokers are the sites that help consumers to filter the required information from a myriad of publicly available information sources. For example, an information broker site will help you to find the site that sells the book you look for to purchase at the lowest price[18.] This helps consumers to save their valuable time searching for information and doing comparisons based on multiple criteria manually. Information brokering sites utilizes agent technology for the brokering process.

- **Content providers:** Content providers are the sites that offer digital content such as news, music, photos, videos and games online. One of the best examples for a content provider is Apple Inc.'s iTunes Store which revolutionized the music industry of the world. Content providers make money by charging for downloads, providing display space for advertisements or subscription fees[19].

- **Portals:** Portals provide the entry point to e-commerce by aggregating data from a large number of providers and letting customers search for goods and services[20]. Most modern portals not only provide the searching facility but also provide the facilities for personalizing the content based on user profile make recommendations on products and services and deliver personalized advertisements[21]. The search engine www.yahoo.com is a very good example of an online portal through which the users can reach numerous categories of products and services.

- **Online service providers:** Online service providers provide various services online such as news, search engine, banking, health care, entertainment, social networking, etc. Facebook, for example, provides social networking services to worldwide users online. Google, on the other hand, provides various online services including search engine, email, file storage, and social networking.

- **Virtual communities:** Virtual communities are formed by individuals connected through online social networks sharing similar interests, opinions or feelings[i]. For example, the fan

[18] Read the example on http://www.cs.toronto.edu/km/xib/document/broker_tutorial/definition.html
[19] Read the article http://ecommercetoyouu.blogspot.com/2015/04/content-provider.html
[20] Read the article https://www.ecommercetimes.com/story/61955.html for examples
[21] Read http://smallbusiness.chron.com/portal-business-model-3869.html

pages and groups on Facebook are virtual communities which comprise of individuals who have similar interests. Virtual communities play an important role in e-commerce as potential pools of customers as well as an effective means of marketing. Virtual communities can also provide resources to firms online in the form of opinions, insights, expertise as well as creative skills.

E-commerce can be classified into several classes such as B2B, B2C, C2B, C2C, B2E, and G2C. This classification is done based on the two parties involved.

- **B2B:** Business to Business (B2B) e-commerce happens when a business organizations purchase goods or services from another business organization online. For example, large manufacturing organizations buy rawmaterials and accessories required for production from suppliers online[22.]
- **B2C:** Business to Consumer (B2C) is the most common form of e-commerce, which means the online buying and selling transactions happening between businesses and their end-customers. Individuals purchasing electronic items online or customers using e-banking services are examples of B2C e-commerce.
- **C2B:** Consumer to Business (C2B) involves consumers offering services (or any kind of resource) to businesses online. For example, some consumers write blog posts on behalf of business organizations after consuming their products which attracts more customers to the firm. Reverse auctions where the consumers name the price for a product or a service is another example for C2B e-commerce[23].
- **C2C:** Consumer to Consumer (C2C) e-commerce is a type of commerce that happens between two online consumers where one sells a product or a service to the other on a dedicated virtual platform. The popular online auction site eBay.com is a renowned example for such a virtual platform dedicated for C2C e-commerce.
- **B2E:** Business to Employee (B2E) e-commerce happens when business firms offer online services to their employees. The objective of B2E is to attract and retain high-quality employees to the organization by offering them various incentives such as flexible working hours, training opportunities and otherbenefits. Businesses typically use online portals to enable B2E e-commerce[24]. Employees can log into the portal using their usernames and passwords and consume the services offered to them through the portal.
- **G2C:** Government to Citizen (G2C) means governments offering various services to their citizens through the Internet. For example, the Office of the Provincial Commissioner of Motor Traffic (Western Province) in Sri Lanka offers the online facility of obtaining revenue licenses for motor vehicles to the citizens[25.] This type of e-commerce has become popular with e-government initiatives around the world.

As mobile devices get proliferated, more and more people browse the Internet using mobile and other handheld devices. As such, firms have now gone beyond traditional e-commerce and have focused on mobile-based e-commerce transactions which are commonly known as m-commerce. Mobile-friendly web sites, as well as mobile based information services such as geo-advertising and geo-information services, are getting widespread making m-commerce the next generation electronic commerce.

[22] Read https://www.linkedin.com/pulse/worlds-12-b2b-websites-currently-holds-top-ranks-saqib-ilyas/
[23] Read https://www.businessnewsdaily.com/5001-what-is-c2b.html
[24] Read https://searchcio.techtarget.com/definition/B2E
[25]https://www.erevenuelicense.motortraffic.wp.gov.lk/erl/view/logout.action

ICT has also contributed largely to creating more secure and alternative payment mechanisms for business transactions. Especially in electronic commerce, several alternative payment mechanisms are preferred over traditional cash-based payment methods such as cash on delivery (COD). One of the key ICT-based developments in payment mechanisms is the payment gateways which enable payments using credit/debit cards. A payment gateway is a software service that securely handles the credit card information exchange and verification in order to complete an electronic commerce transaction. Once the customer authenticated a credit card payment for an online purchase he/she wishes to proceed, the merchant's webserver securely collects the credit card details (in encrypted format) from the customer's browser and sends the details of the payment to its payment gateway. The payment gateway directs the details to the payment processor of the merchant's bank through which the details will be sent to the relevant card association (Visa, MasterCard, etc.) The card association directs the details to the card-issuing bank which responses with an accept/reject message after a verification process. The processor forwards the response to the payment gateway to be interpreted at the merchant's website. If accepted, the transaction gets completed and the customer gets a confirmation message, whereas if rejected, the transaction gets aborted delivering a failure message to the customer. The merchant submits all approved payments to the processor of his/her bank to process as a batch to get money transferred to the designated account[26].

One issue with online payments with payment gateways is the customer has to submit credit card details to every merchant (or rather at every merchant's web site) that they shop with. This increases the number of chances of getting your sensitive data being recorded by one of those merchants. There are third party systems such as PayPal that reduces this risk by mediating the payment process and allow sending and receiving money more securely. In the case of PayPal, both buyers and sellers can register with PayPal using a valid email address and link any number of bank accounts and credit cards to that. Buyers can then shop securely with sellers who allow payment with PayPal (usually those who have the 'Checkout with PayPal' option). A buyer who makes a payment through PayPal can choose the bank account or the credit card he/she wishes to use to make the payment. PayPal does not disclose any sensitive information of the buyer to the seller but mediates the payment authentication process and ensures that the seller receives money to the designated bank account in his/her PayPal account[27].

Virtual currencies, in particular, the cryptocurrencies are also getting popular as alternative payment mechanisms in online payments. A cryptocurrency is a digital equivalent to cash which could be used to exchange for goods and services. Cryptocurrencies use cryptography to control the generation of additional units as well as to secure transactions. Among the popular cryptocurrencies, there are Bitcoin, Litecoin, Ethereum, and Zcash. The cryptocurrencies work based on the technology called 'blockchain' and individuals can buy/sell cryptocurrencies through third-party websites (and their mobile apps) using credit or debit cards. However, some countries have fully banned or restricted the usage of cryptocurrencies. The central bank of Sri Lanka is also closely paying attention to the cryptocurrency-related activities happening in the country[28].

Apart from security, privacy is another challenging issue in e-commerce. E-commerce sites keep track of personal information of their visitors. Moreover, they monitor what customers are doing online by tracking what sites they visit, what products and services they search as well as what products and services do they buy. Using this information, they make profiles of customers and

[26] Read the article on https://www.hostmerchantservices.com/articles/payment-gateway-articles/how-payment-gateways-work/

[27] Watch the video on https://www.youtube.com/watch?v=7aFqCqqaBZQ&t=230s

[28] https://www.cbsl.gov.lk/en/news/public-awareness-on-virtual-currencies-in-sri-lanka

accordingly conduct targeted promotions. However, customers may not always be aware to what extent have they been monitored and there is a growing concern among customers about their privacy getting violated. Hence, Government agencies work on insisting e-commerce sites to have privacy policies to ensure that the customers' privacy is protected at their sites[29].

ICT in the Forefront

ICT is widely used by companies to connect with their customers and promote their products and services. The Internet and other electronic media provide a range of technologies and platforms to communicate a strong message to the customers regarding a firm's offering.

[29] Read the article https://cyber.harvard.edu/olds/ecommerce/privacytext.html

Competency Level 12.3 Analyses the ICT in terms of generating and delivering an improved products and services to consumers

Learning Outcomes:

- Defines e- marketing
- Identifies the role of ICT in e-marketing
- Investigates the usage of databases in marketing activities to improve the product and services according to the requirements of the customers
- Identifies the ways of gaining competitive advantages using ICT

E_Marketing

Electronic marketing is also known as web marketing, Internet marketing, as well as online marketing. It is simply making use of electronic media, especially the Internet for marketing activities[30]. Such activities typically include promotions, public relations as well as after-sales support. For example, many customers make purchases based on the emails they receive containing product catalogs and special offers. Social media that are being increasingly used by firms to stay connected with customers, share information regarding new offerings and promotions as well as to hear from customers any suggestions, complaints, as well as comments are some of the other examples. Social media, on the other hand, acts as the digital equivalent to word-of-mouth marketing as the customers help to spread the message about the firm's offerings by sharing, tagging, liking and commenting. Online advertisements are also very effective to spread the message about a particular product or a service. The advantages of e-marketing include the ability of a firm to adapt based on the customers' feedbacks, the ability to make an instant impact, the ability to directly approach specific market niches (based on profiling), the ability to analyze the responses in real-time, etc.[31]

Mobile Marketing

Mobile marketing is a revolutionary form of e-marketing. As we know, many people now use smartphones and other mobile handheld devices to browse the Internet. This shift from desktops to mobile devices has created new opportunities for marketers to use multiple channels to reach a customer through their mobile handheld devices. These channels may include email, SMS, MMS as well as the very popular social media such as Facebook, Viber, WhatsApp, etc. Technology is developing rapidly to create innovative ways to personalize the marketing messages sent to users as much as possible. According to the latest edition of the textbook - Management Information Systems by Laudon and Laudon – the proliferation of mobile devices introduces a new dimension to marketers to recognize their targets, which is the location. They introduce three notable location-based services the mobile marketers can use namely geosocial services, Geoadvertising and geoinformation services. The geosocial services help users to know where people with similar interests as him or her meet. Facebook is a popular social networking site that offers this service. Geoadvertising services help connecting local merchants with nearby customers. For example, the local restaurants can offer lunchtime promotions to the crowd within a specific distance expecting to attract a percentage of them as customers. Geoinformation services provide information about a subject, such as an exhibit at a museum, on a geographical map for the people passing nearby.

[30] Note: marketing is not only promotion or advertising

[31] Read http://www.iaapa.org/docs/handout-archive---ops/Mon_KHAN_E-MARKETING.pdf

M-commerce is an important component of mobile marketing in which electronic commerce takes place through mobile devices. In other words, the customers today prefer to shop online for products and services through their mobile devices which makes firms to consider the mobile as an important channel to transact with customers. For example, most banks, insurance firms, and telco companies offer mobile apps for customers to receive their services through mobile devices. More and more firms offer mobile-friendly e-commerce sites expecting better engagement with the customers. There also exist mobile advertising platforms such as Apple's iAd and Google's AdMob.

Database Marketing

When the customers' online activities increase through both web and mobile interfaces, more and more databases become existing which contain information about current and prospective customers. These databases not only contain customers' personal information but also their web browsing patterns as well as location details. Database marketing involves making use of these databases to directly contact existing or potential customers regarding firms' offerings. Database marketing is a direct marketing technique and it could be well supported with intelligent data mining techniques. For example, mining of web access patterns of individual customers can help to create unique profiles of customers based on their interests and other characteristics. This helps to make more personalized offers to customers which are more likely to be accepted than a bulk offer sent to all customers irrespective of their likes and dislikes.

Review questions

1. Explain what is digital transformation
2. Discuss the digital transformation happened in various business fields using the modern taxi apps and the websites like Airbnb and Amazon
3. Explain how information technology becomes important during global emergency situations such as pandemics
4. Differentiate electronic business and electronic commerce
5. Write down different online payment methods
6. Explain the functionality of an online payment gateway using a suitable diagram
7. Explain two new examples for each of the e-commerce types described above
8. Explain how the Internet of Things, Big Data and Cloud Computing technologies could be used to create a Smart System
9. Explain how social media could be effectively used for businesses
10. Explain why the smart mobile phone has become an important tool in business
11. Briefly describe the functionality of an information system developed for human resource management
12. Discuss whether the famous website 'YouTube' an example for a content provider